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18.01 Single Variable Calculus
Fall 2006

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Practice Final SOLUTIONS 18.01

1 a) $\dfrac{x^2\cdot\frac{1}{x}-\ln x\cdot 2x}{x^4}=\dfrac{1-2\ln x}{x^3}$

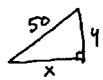
b) $\frac{1}{2}(3\sin^2u+2)^{-1/2}\cdot 6\sin u\cos u$
$=\dfrac{3\sin u\cos u}{\sqrt{3\sin^2u+2}}$

c) $D^n e^{kx}=k^n e^x$; at 0: $\boxed{k^n}$

2 $D_x(x^2y^2+y^3)=2xy^2+x^2\cdot 2yy'+3y^2y'$
$=0.$
At (1,1): $2+2y'+3y'=0$; $y'=-\frac{2}{5}$
Eq'n: $(y-1)=-\frac{2}{5}(x-1)$ or
$y=-\frac{2}{5}x+\frac{7}{5}$

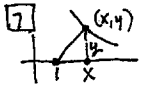
3 $y=\cos^{-1}x$ ↗ $y'=\dfrac{-1}{\sin y}=\dfrac{-1}{\sqrt{1-\cos^2y}}$
$x=\cos y$
$1=-\sin y\cdot y'$
$\therefore\ y'=\dfrac{-1}{\sqrt{1-x^2}}$
(use $+\sqrt{}$ since $\sin y>0$)

4 $\begin{cases}x^2+x+a\\ bx+2\end{cases}$ same value at 0 (for continuity) $\Rightarrow \boxed{a=2}$
same deriv. at 0 (for diff'y) $\Rightarrow 2x+1=b$ at 0
$\therefore\ \boxed{b=1}$

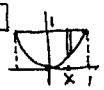
5


$x^2+y^2=50^2$
$2x\frac{dx}{dt}+2y\frac{dy}{dt}=0$
when $y=30$, $x=40$: $40\cdot\frac{dx}{dt}+30(-2)=0$
$\frac{dx}{dt}=\frac{60}{40}=\boxed{\frac{3}{2}}$ ft/sec

6 $A=2x\cdot(1-x^2)=2x-2x^3$
$\frac{dA}{dx}=2-6x^2=0$ if $x^2=\frac{1}{3}$
$x=\frac{1}{\sqrt{3}}$
Area then is:
$2\cdot\frac{1}{\sqrt{3}}\left(1-\frac{1}{3}\right)=\boxed{\frac{4}{3\sqrt{3}}}$

7


$-\frac{1}{y'}=\frac{y}{x-1}$ (negative recip. slope)
a) $\therefore\ y'=\frac{1-x}{y}$
b) $\frac{dy}{dx}=\frac{1-x}{y}$
$y\,dy=(1-x)\,dx$ or better: $\left[\frac{(1-x)^2}{-2}+c_2\right]$
$\frac{1}{2}y^2=x-\frac{1}{2}x^2+c_1$
$y^2=2x-x^2+c$
c) $y^2+(x-1)^2=c_3$ (completing squares)
cuides centered at (1,0)

8


Volume $=\int_0^1 2\pi x(1-x^2)\,dx$
$=2\pi\left(\frac{x^2}{2}-\frac{x^4}{4}\right)\Big]_0^1$
$=2\pi\cdot\frac{1}{4}=\boxed{\frac{\pi}{2}}$
[or: $\pi\cdot1^2\cdot1-\int_0^1 2\pi x^3\,dx$ = vol. cylinder − vol. under curve $=\pi-\frac{\pi}{2}=\frac{\pi}{2}$]

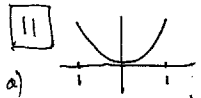
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x	0	$\pi/6$	$\pi/3$	$\pi/2$
$\sin x$	0	$1/2$	$\sqrt{3}/2$	1
$\sin^2x$	0	$1/4$	$3/4$	1



Trap. rule: $\frac{\pi}{6}\left(\frac{1}{2}\cdot0+\frac{1}{4}+\frac{3}{4}+\frac{1}{2}\cdot1\right)$
$=\frac{\pi}{6}\cdot\frac{3}{2}=\boxed{\frac{\pi}{4}}$

10 $F(x)=\int_0^x e^{-t^2}dt$
a) $F'(x)=e^{-x^2}$; $F''(x)=-2xe^{-x^2}$
$F'(1)=\frac{1}{e}$ $F''(1)=\frac{-2}{e}$
b) $\int_1^2 e^{-u^2/4}\,du=\int_{1/2}^1 e^{-t^2}\cdot 2\,dt$
Put $t=u/2$, $dt=\frac{du}{2}$ $=2\left(F(1)-F(\frac{1}{2})\right)$

11

$y=x^2/10$
$y'=x/5$
a) $\therefore$ arclength $=\int_{-1}^1\sqrt{1+\frac{x^2}{25}}\,dx$
b) average $=\int_0^1\frac{x^2}{10}dx=\frac{x^3}{30}\Big]_0^1$
$=\frac{1}{30}$ km ≈ 33 m
[or $=\frac{1}{2}\int_{-1}^1\frac{x^2}{10}dx=\cdots$]

12 $\frac{1}{x^2+3x+2}=\frac{1}{(x+2)(x+1)}=\frac{-1}{x+2}+\frac{1}{x+1}$ (by cover up)
a) $\therefore\int_0^1\left(\frac{-1}{x+2}+\frac{1}{x+1}\right)dx=-\ln|x+2|+\ln|x+1|\Big]_0^1$
$=-\ln3+\ln2+\ln2-0$
$=2\ln2-\ln3$ (or $\ln\frac{4}{3}$)
b) $\int x^2\ln x\,dx=\frac{x^3}{3}\ln x-\int\frac{x^3}{3}\cdot\frac{1}{x}dx$ int. by parts
$=\frac{x^3}{3}\ln x-\frac{x^3}{9}+C$

13 $\int_0^1\frac{dx}{(x^2+1)^2}=\int_0^{\pi/4}\frac{\sec^2u\,du}{\sec^4u}=\int_0^{\pi/4}\cos^2u\,du$
$x=\tan u$
$=\int_0^{\pi/4}\frac{1+\cos2u}{2}du=\frac{u}{2}+\frac{\sin2u}{4}\Big]_0^{\pi/4}$
$=\boxed{\frac{\pi}{8}+\frac{1}{4}}$

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Area $=\frac{1}{2}\int_0^{2\pi}(e^{\theta/2\pi})^2\,d\theta$
$=\frac{1}{2}\int_0^{2\pi}e^{\theta/\pi}\,d\theta=\frac{\pi}{2}e^{\theta/\pi}\Big]_0^{2\pi}$
$=\frac{\pi}{2}(e^2-1)$

15 a) $\lim_{x\to0}\frac{\sin^2x}{1-\cos x}=\lim_{x\to0}\frac{2\sin x\cancel{\cos x}}{\cancel{\sin x}}=2$
b) $\lim_{x\to1}\frac{(\ln x)^2}{x-1}=\lim_{x\to1}\frac{2\ln x\cdot\frac{1}{x}}{1}=0$
c) $\lim_{x\to\infty}\frac{x^2}{e^x}=\lim_{x\to\infty}\frac{2x}{e^x}=\lim_{x\to\infty}\frac{2}{e^x}=0$

16 $\int_1^\infty\frac{dx}{x^{3/2}}=\int_1^\infty x^{-3/2}dx=-2x^{-1/2}\Big]_1^\infty=0-(-2)$
$=2$

17 $\frac{n}{\sqrt{4+n^p}}\sim\frac{n}{n^{p/2}}\sim\frac{1}{n^{p/2-1}}$
$\therefore\ \sum\frac{n}{\sqrt{4+n^p}}$ converges if $\frac{p}{2}-1>1$ or $\boxed{p>4}$

18 $y=(1+x)^{1/2}$ at $x=0$: 1 (coeff: 1)
$y'=\frac{1}{2}(1+x)^{-1/2}$ $\frac{1}{2}$ $\frac{1}{2}$
$y''=-\frac{1}{2}\cdot\frac{1}{2}(1+x)^{-3/2}$ $-\frac{1}{4}$ $-\frac{1}{4}\cdot\frac{1}{2!}$
$y'''=-\frac{3}{2}\cdot-\frac{1}{2}\cdot\frac{1}{2}(1+x)^{-5/2}$ $\frac{3}{8}$ $\frac{3}{8}\cdot\frac{1}{3!}$
$\therefore(1+x)^{1/2}=1+\frac{1}{2}x-\frac{1}{8}x^2+\frac{1}{16}x^3$
$(1.2)^{1/2}=1+\frac{1}{2}\cdot\frac{2}{10}-\frac{1}{8}\cdot\frac{4}{100}+\frac{1}{16}\cdot\frac{8}{1000}$
$=1+.1000-.005+.0005$
$=\boxed{1.0955}$

19 $y=\tan^{-1}x$
$y'=\frac{1}{1+x^2}=1-x^2+x^4-x^6+x^8\ldots$?
integrate term by term: $\therefore\ y=\tan^{-1}x=x-\frac{x^3}{3}+\frac{x^5}{5}-\frac{x^7}{7}+\frac{x^9}{9}\ldots+C$
$C=0$ since $\tan^{-1}0=0$.
all terms including x, so f(0) = 0, no need c